FIN 412: THEORY OF CORPORATE FINANCE

CAPITAL MARKETS, CONSUMPTION AND INVESTMENT DECISION

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COURSE RESOURCES

- **The lecture is based on the recommended text for the course**
- **Main reading:**

- Olowe, R. A. (2017). Financial Management: Concepts, Financial System and Business Finance. 4th edition. Lagos: Brierly Jones (ORA)
- Copeland T., and Weston J. (1983). Financial Theory and Corporate Policy. Latest edition. (CTW)
- Brealey/ Myers. Principles of Corporate Finance.10th Edition (or earlier). (BM)
- Frantz/Payne/Favilukis. Study Guide. Corporate Finance. First Edition. 2011. (FPF)

COURSE RESOURCES

- **Supplementary reading**:

- Ross S., R.Westerfield, J.Jaffe. (1999). Corporate Finance. Fifth Edition. IRWIN-McGraw-Hill.
- Damodaran A. (1999). Applied Corporate Finance. Wiley&Sons.
- Daniel, K., & Marshall, D. (1997). The equity premium puzzle and the risk-free rate puzzle at long horizons'. Macroeconomic Dynamics l, 452-484.
- DeBondt, W., & Thaler, R. (1985). Does the Stock Market Overreact?, *Journal of Finance*, 40, 793-805.
- Dimson, E., & Mussavian, M. (2000). Market Efficiency, The Current State of Business Disciplines, vol.3, Spellbound Publications, 2000, pp 959-970
- Fama, E.F. (1998). Market Efficiency, Long-Term Returns, and Behavioral Finance. *Journal of Finance*, 49(3) 283-306.
- Fama, E. F., & French, K. (1992).The Cross-Section of Expected Returns, *Journal of Finance*, 47, 427- 465.
- Fama, E. F. (1970). Efficient Capital markets: A Review of Theory and Empirical Work, *Journal of Finance*, 25 (2), 383-417

CONTENT

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- Consumption and Investment Decision with Production and Capital Markets
- Fisher Separation Theorem
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Overview
(CTW – Chap. 1)

- Consumption and investment decisions made by individuals and firms will help us to appreciate the role of interest rates in making making these decisions.
- The simplest of all worlds, a one-person/one-good economy with no uncertainty, the Robinson Crusoe world.
- The decision maker, Robinson Crusoe, must choose between consumption now and consumption in the future.
- The decision not to consume now is the same as investment.
- Thus Robinson Crusoe's decision is simultaneously one of consumption and investment.

Consumption and Investment Decision without Capital Markets(CTW – Chap. 14)

- We start the discussion with our one man one good economy . Robin Crusoe economy.
- Robin Crusoe needs to take a decision between consumption now and consumption in the future
- Two types of information are need

➢ subjective trade-offs between consumption now and consumption in the future. This is embedded in the Utility and indifference curves (Figure 1)

➢ investment and production opportunity sets (Figure 2). This shows the feasible trade-offs between present and future consumption that are technologically possible.

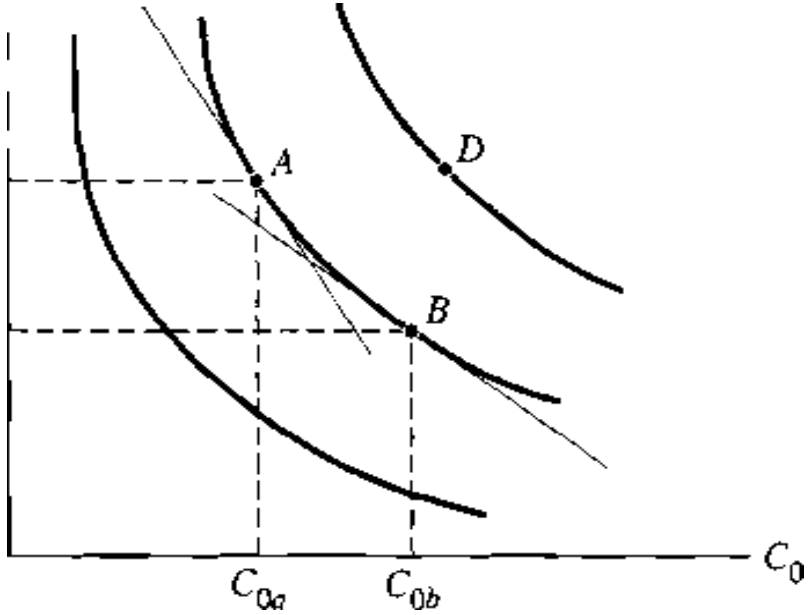
Consumption and Investment Decision without Capital Markets - Assumptions

- We will work with the following assumptions

- Outcome from investment are known with certainty
- There are no transaction cost or taxes
- Decision are made in one-period context.
- Individuals are endowed with income at the beginning of the period and at the end of the period. They must decide how much to consume now and how much to invest.
- Every individual is assumed to prefer more wealth to less i.e. marginal utility of consumption is always positive.
- The marginal utility of consumption is decreasing.

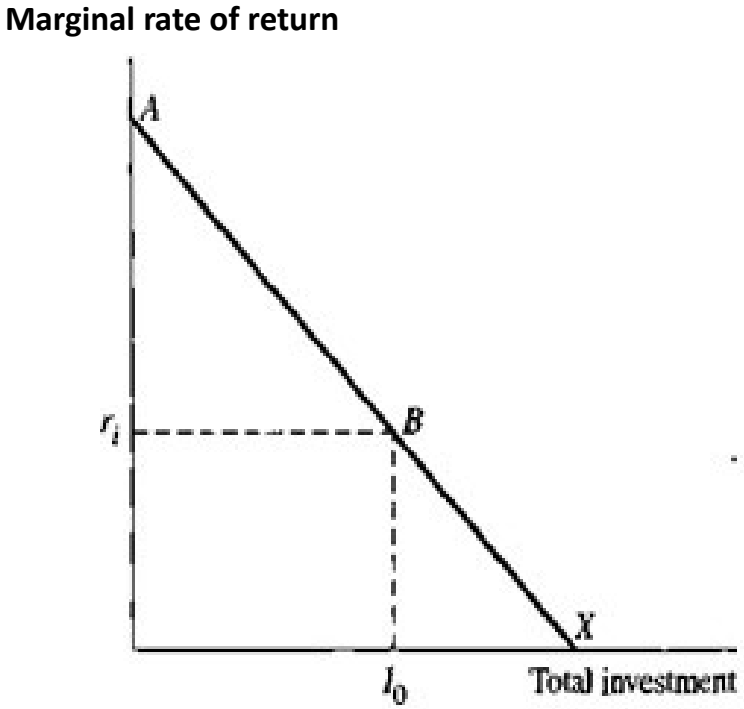
Consumption and Investment Decision without Capital Markets - Assumptions

- **Figure 1** Indifference curves representing the time preference of consumption (CTW)



Consumption and Investment Decision without Capital Markets

Figure 2 Individual's Schedule of Productive Investment Opportunities (CWT)



Consumption and Investment Decision without Capital Markets

- Figure 1 shows the preference functions telling us how individuals make choices among consumption bundles over time.
- The slope of the indifference curve in Figure 1 is called Marginal rate of substitution (MRS). It is also called subjective rate of time preference -(1+r).

$$\text{MRS} = \frac{\partial C_1}{\partial C_0} = -(1+r)$$

- The subjective rate of time preference is greater at point *A* than at point *B* due to the convexity of the indifference curve.
- At point A, the individual consume less today and demand relatively more future consumption in order to have the same total utility.

Consumption and Investment Decision without Capital Markets

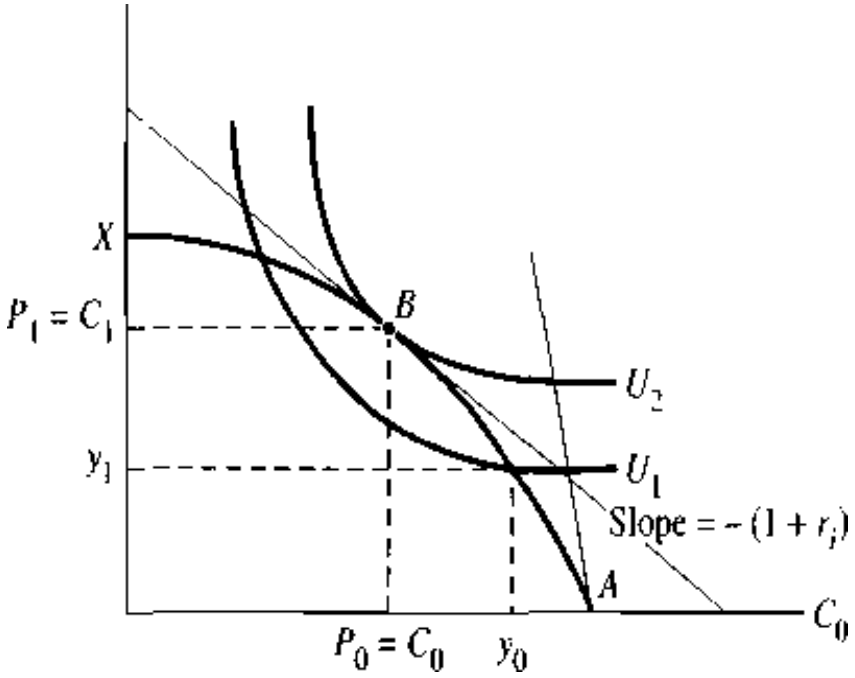
- Figure 2 shows schedule of productive/investment opportunities arranged from the highest rate of return down to the lowest that allow a unit of current savings/investment to be turned into more than one unit of future consumption
- Practically, the investment opportunities graph of Figure 2 though shown as a straight line is a curve with decreasing function. This is because of diminishing marginal returns to investment.
- The more an individual invests, the lower the rate of return on the marginal investment.
- All investments are assumed independent of one another and perfectly divisible

Consumption and Investment Decision without Capital Markets

- An individual will make all investments in productive opportunities can be better demonstrated if we combine graphs of Figures 1 and 2 as shown in graph of Figure 3.
- As shown in Figure 3, a rational individual will invest in productive opportunities with higher rates of return than his or her subjective rate of time preference
- The slope of a line tangent to curve *ABX* in Fig. 1.5 is the rate at which a dollar of consumption foregone today is transformed by productive investment into a dollar of consumption tomorrow. It is the *marginal rate of transformation* (MRT) offered by the production/investrnent opportunity set.

Consumption and Investment Decision without Capital Markets

Figure 3 Production Opportunity Set (CWT)



Consumption and Investment Decision without Capital Markets

- In Figure 5, Curve ABX is the Productive opportunity set
- The slope of a line tangent to curve *ABX* in Figure 3 is called the *marginal rate of transformation* (MRT) offered by the production/investment opportunity set.
- MRT is the rate at which a naira of consumption foregone today is transformed by productive investment into a naira of consumption tomorrow.
- Point A has the highest slope in Figure 3 and represents the highest rate of return at point A in Figure 3

Consumption and Investment Decision without Capital Markets

- We will start the point of initial endowment of the individual (y0,y1). The individual decision maker compares the marginal rate of return on a naira of productive investment (or disinvestment) with his or her subjective time preference.
- If the rate on investment is greater he or she will gain utility by making the investment.
- This process continues until the rate of return on the last naira of productive investment just equals the rate of subjective time preference (at point *B).*
- Thus, at point B, the investor's subjective marginal rate of substitution is equal to the marginal rate of transformation offered by the production opportunity set, that is:

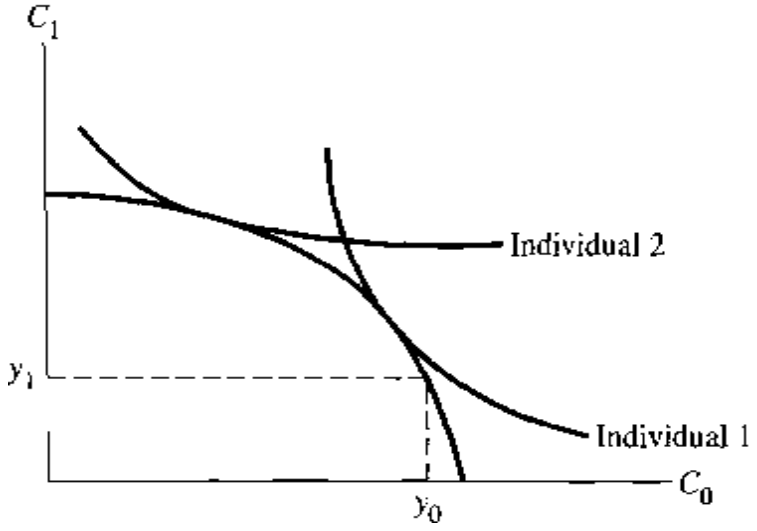
$$MRS = MRT$$

Consumption and Investment Decision without Capital Markets

- Point B is the point of optimal consumption decision. The individual will C0 today and C1 tomorrow. The individual will invest y0 – C0 today.
- The individual gained utility by moving from lower level indifference U1 tyo a higher level U2.
- Further, at point *B* the individual's consumption in each time period is exactly equal to the output from production (i.e. P0 = C0 and P1 = C1).
- The solution as shown in Figure 3 applies to only onew individual (Robin Crusoe)

Consumption and Investment Decision without Capital Markets – With more than One Individuals

- Individuals with the same endowment and the same investment opportunity set may choose completely different investments because they have different indifference curves as shown in Figure 4.
- **Figure 4**: Individuals with different indifference curves choose different production consumption patterns.



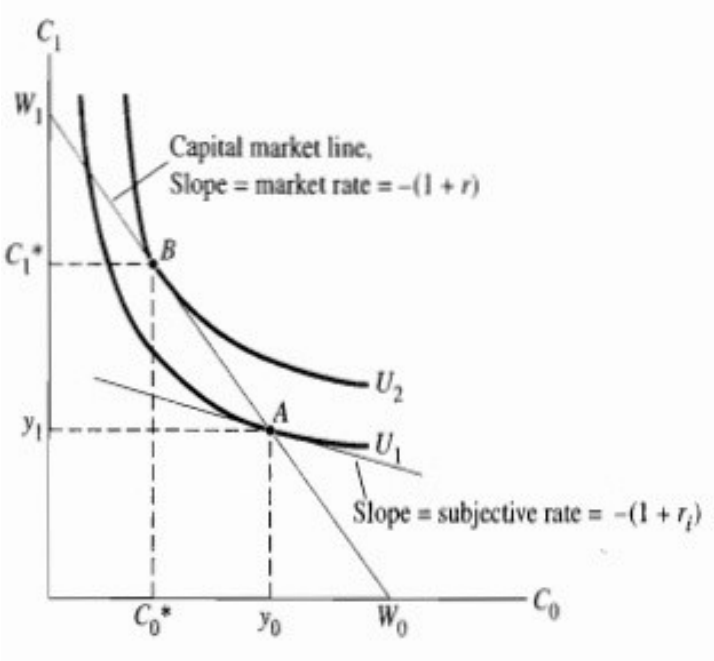
Consumption and Investment Decision with Capital Markets

- In a Robinson Crusoe economy, there are no opportunities to exchange intertemporal consumption among individuals.
- The introduction of Capital markets facilitate the Intertemporal exchange of consumption represented by the opportunity to borrow or lend unlimited amounts at r, a market-determined rate of interest
- Capital markets facilitate the transfer of funds between lenders and borrowers. Money lent today (X_0) will return principal plus interest at the end of the period (X_1)

  $X_1 = X_0(1+r)$.
- The capital market line as represented by W_0ABW_1 in Figure 5 shows borrowing and lending opportunities

Consumption and Investment Decision with Capital Markets

Figure 5: The Capital Market Line and Consumption



Consumption and Investment Decision with Capital Markets

- At A in Figure 5, the point of initial endowment, a rational individual will desire to lend because the subjective time preference is less than the market rate of return offered by the capital market.
- Utility will be maximized by moving along the market line to Point B where the subjective time preference equals the market interest rate. Point *B* represents the consumption bundle on the highest attainable indifference curve.

Consumption and Investment Decision with Capital Markets

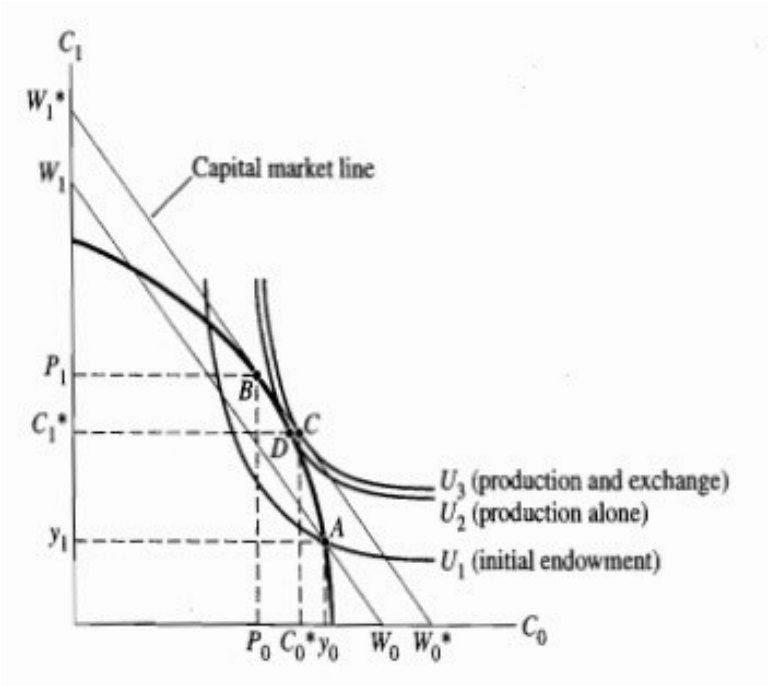
- Thus, the capital market line as shown in Figure 4 has an intercept at W_1 and a slope of -(1 + r).
- Equating the two current wealth (W_0) equations show that the present value of endowment equals the present value of consumption
- Thus, movement along the capital Market Line does not change an individual's wealth, but offer a pattern of consumption that has higher utility.

Consumption and Investment Decision with Production and Capital Markets

- The analysis of consumption and investment decision in Figure 5 ignores production
- Let us combine production opportunity set with Figure 5 so that we can see how production/consumption decision takes place in a world where capital markets facilitate the exchange of funds at the market rate of interest
- Figure 6 combines production possibilities with market exchange possibilities.
- In Figure 6, there are 3 indifference curves, U_1, U_2 and U_3 and we also have initial endowment (y_0, y_1) at A.

Consumption and Investment Decision with Production and Capital Markets

Figure 6: Production and Consumption with Capital Markets



Consumption and Investment Decision with Production and Capital Markets

- Just as in Figure 4, we will start the discussion at point A , the point of initial endowment (y_0, y_1).
- At point A, movement can be either be along the production opportunity set or along the capital market line as both alternatives offer a higher rate of return than the subjective time preference. However, production offers the higher return (i.e., a steeper slope). Thus, a rational individual will invest and move along the production opportunity frontier.
- .Without the capital market line that facilitated the opportunity to borrow or lend, investment will stop at point D, where the marginal rate of return on productive investment equals the subjective time preference. This was the result shown for consumption and investment in a Robinson Crusoe economy in Figure 3. At point D, the level of utility has increased from U_1 to U_2.

Consumption and Investment Decision with Production and Capital Markets

- In Figure 6, investment will not stop at Point D since at that at point the borrowing rate, represented by the slope of the capital market line, is less than the rate of return on the marginal investment, which is the slope of the production opportunity set at point D. Thus, the individual will earn more returns on further investment than the borrowing cost.
- Investment will continue until marginal return on investment is equal to the borrowing rate at point B.
- At point B, the individual will receive output from production (P_0, P_1). The present value of wealth is now W_0*instead of W_0. The individual can now reach any point on the market line. However, as can be seen from Figure 6, time preference at point B is greater than the market rate of return, implying that an individual can consume more than P_0, the current payoff from production.

Consumption and Investment Decision with Production and Capital Markets

- With borrowing, the individual can move along the capital market line until it reaches point C, where the subjective time preference just equals the market rate of return. This is the optimal consumption decision.
- It should be noted that utility has increased from U_1 at point A (initial endowment) to U_2 at point *D* (the Robinson Crusoe solution) to U_3 at point *C* (the exchange economy solution).
- Thus, clearly the capital market is better off for the economy as U_3 is higher than U_2

Consumption and Investment Decision with Production and Capital Markets – Fisher Separation Theorem

- Thus, with production opportunities and capital market exchange opportunities, the decision process takes place occurs in two separate and distinct steps:

➢ Step 1: choose the optimal production decision by investing on projects until the marginal rate of return on investment equals the objective market rate;

➢ Step 2: choose the optimal consumption pattern by borrowing or lending along the capital market line to equate the subjective time preference with the market rate of return.

- Thus, there is a separation of the investment and consumption decisions and is known as the **Fisher separation theorem.**

Consumption and Investment Decision with Production and Capital Markets – Fisher Separation Theorem

- **FISHER SEPARATION THEOREM**

- *Given perfect and complete capital markets, the production decision is governed solely by an objective market criterion (represented by maximizing attained wealth) without regard to individuals' subjective preferences that enter into their consumption decisions*.

Implications of Fisher Separation Theorem

- With the Fisher Separation theorem, there would be unanimity among the shareholders/investors as to the production decision for a Firm.
- Consequently, investment decision can be delegated to managers by the shareholders regardless of the shape of their indifference curves.
- The Shareholders can then take the output of the firm and adapt it to their own subjective time preferences by borrowing or lending in the capital market (CWT).

Transaction Cost and the Breakdown of Separation

- As shown in Figure 8, individuals will not be able to delegate the investment decision to the manager of their firm without a single market rate.
- Figure 8 shows that Individual 1 would direct the manager to use the lending rate and invest at point B while Individual 2 would use the borrowing rate and choose point A.
- Despite this imperfection, it should be noted that the theory of finance is based on the assumption of perfect capital markets. Although not realistic (empirical issue), several practical applications have been developed based on this assumption. Throughout the rest of our discussions in this course, we shall work with the assumptions of perfect capital market

SUMMARY

- Consumption and investment decisions made by individuals and firms will help us to appreciate the role of interest rates in making making these decisions.
- In a Robin Crusoe Economy, optimal consumption takes place at the point where the investor's subjective marginal rate of substitution is equal to the marginal rate of transformation

SUMMARY

- In a world of Capital market that facilitate exchange possibilities, there is a separation of production decision and consumption decision. Optimal Production decision will occur by investing on projects until the marginal rate of return on investment equals the objective market rate. Optimal consumption will take place by borrowing or lending along the capital market line to equate the subjective time preference with the market rate of return.
- Utility is higher with capital markets compared with Robinson Crusoe
- The economy is better off with capital markets

Review Question

- Graphically discuss the Fisher Separation Theorem
- Suppose that the investment opportunity set has M projects, all of which have the same rate of return, R. Graph the investment opportunity set